Arcs complete outside a prescribed conic

A prescribed-hole defect identity and a certified classification over $\mathbb{F}_{16}$

Tavis Rudd

July 2026

Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$. We study arcs $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Starting from the classical first two equations for the secant-index distribution of an arc, we prove an exact defect identity for an arbitrary prescribed set of points that may remain uncovered, together with equality and stability criteria. Its conic specialization gives

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{\lfloor k/2 \rfloor}$$

for every $\mathcal{C}$-complete k-arc, together with an exact equality criterion and a quantitative stability estimate. In particular, for every prime power q,

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

Conversely, if H is any prescribed set and an ordinary complete b-arc satisfies $b|H| < q^2 + q + 1$, projective averaging moves it off H; the conic case gives $\rho_{\mathcal{C}}(q) \leq t_2(2,q)$ whenever $t_2(2,q) \leq q$. We also give tangent and parity constraints in even characteristic. Directly checkable witnesses give

$$\rho_{\mathcal{C}}(8) = \rho_{\mathcal{C}}(9) = \rho_{\mathcal{C}}(11) = 6, \qquad \rho_{\mathcal{C}}(16) = 9.$$

A classification of all 2633 projective eight-arcs over $\mathbb{F}_{16}$ proves the stronger statement that no nonzero quadratic can contain an eight-arc's ordinary-uncovered locus while avoiding the arc. A supplementary verifier checks the witnesses, while a companion Lean formalization proves the theorem chain and kernel-checks the local classification, quadratic obstruction, projective reduction, and exact $q = 16$ conclusion.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [3], Kim–Vu [8], and Nagy [9].

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2,q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ is *$\mathcal{C}$-complete* if every point of

$$\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

For an arbitrary arc disjoint from $\mathcal{C}$, its uncovered locus is

$$\mathcal{U}_{\mathcal{C}}(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \mathcal{C} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

Thus A is $\mathcal{C}$-complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$.

The minimum in Definition 1.1 is attained. Indeed, in the finite family of arcs disjoint from $\mathcal{C}$, any member of maximum cardinality is maximal under insertion and hence is $\mathcal{C}$-complete. The same argument works for every prescribed hole set.

Equivalently, A is maximal among arcs in the point set $\mathrm{PG}(2,q) \setminus \mathcal{C}$. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

The problem differs from two nearby standard notions. A saturating set need not be an arc and has no prescribed exceptional locus. An almost-complete subset of a conic, in the sense of Bartoli–Davydov–Marcugini–Pambianco [5], consists of selected points *on* the conic whose chords cover the complementary points. Ng–Wild's arcs covering a line [10] prescribe a one-dimensional set that must be covered, rather than an exceptional set that may remain uncovered. Here the selected arc is required to avoid the conic.

The point index with respect to an arc, and the first two double-counting equations for its distribution, are classical; compare Hirschfeld [7, Chapter 9] and Ball [3]. The use of secant counts to bound the smallest complete (k,n)-arcs is also classical; for a recent explicit lower bound see Alabdullah–Hirschfeld [1]. The main observation of this note is that, after splitting those equations over a prescribed exceptional set and its complement, the inequalities normally used to control overlap have an exact remainder. This produces:

(i) an exact prescribed-hole defect identity and equality criterion;

(ii) a quantitative stability bound for near equality;

(iii) a strengthened lower bound with additive asymptotic term $3/2$;

(iv) a projective-averaging upper-bound transfer from ordinary complete arcs;

(v) characteristic-two nucleus constraints; and

(vi) small finite-field results whose upper bounds are independently verifiable from explicit coordinates.

A targeted literature search across complete and saturating arcs, almost-complete conic subsets, prescribed-line coverage, arcs and quadrics, and the published classification of arcs in $\mathrm{PG}(2,16)$ did not locate Definition 1.1 or the exact defect identity below in this form. This observation is not a priority certificate: the proofs are elementary consequences of classical index equations, and terminology involving relative saturation or holes may occur elsewhere.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q, let A be a k-arc in Π, and let $r_A(x)$ be the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A. Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A, of which there are at most $\lfloor k/2 \rfloor$. □

Proposition 2.2 (Classical first and second index equations)**.** *For every k-arc A in a projective plane of order q,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A, and therefore contributes $q-1$ incidences with points outside A. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. □

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered, and define

$$\begin{aligned} V_{\mathcal{H}}(A) &= \Pi \setminus (A \cup \mathcal{H}), \\ \mathcal{X}_{\mathcal{H}}(A) &= \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}, \\ \mathcal{U}_{\mathcal{H}}(A) &= V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A), \\ I_{\mathcal{H}}(A) &= \sum_{y \in \mathcal{H}} r(y). \end{aligned}$$

Thus $\mathcal{X}_{\mathcal{H}}(A)$ is the covered required locus, and $\mathcal{U}_{\mathcal{H}}(A)$ is the uncovered required locus.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x\in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x\in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y\in\mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

Theorem 3.1 (Prescribed-hole defect identity)**.** *Let A be a k-arc in a projective plane of order q, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y\in\mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y\in\mathcal{H}} \binom{r(y)}{2}.$$

Substitution and termwise simplification yield

$$\begin{aligned} m(r-1) - 2\binom{r}{2} &= (r-1)(m-r), \\ (m-1)r - 2\binom{r}{2} &= r(m-r), \end{aligned}$$

which proves (5). Every term is nonnegative by Lemma 2.1. $\square$

Corollary 3.2 (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \leq N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \geq |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \geq 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). $\square$

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes). *Let $|\mathcal{H}| = h$. If A is a k-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2+q+1-k-h$. Substitute this equality into (6). □

The exact remainder also quantifies near equality.

Corollary 3.4 (Quantitative stability). *Assume $m \ge 3$ and put*

$$\begin{aligned} M &= \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \le r(x) \le m-1\}, \\ J &= \{y \in \mathcal{H} : 1 \le r(y) \le m-1\}. \end{aligned}$$

Then

$$(m-2)|M| + (m-1)|J| \le m\Delta_{\mathcal{H}}(A). \tag{9}$$

Proof. For $2 \le r \le m-1$, $(r-1)(m-r) \ge m-2$. For $1 \le r \le m-1$, $r(m-r) \ge m-1$. Apply these estimates to (5). □

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m, while almost every hole point met by a secant also has multiplicity m. This is the precise stability content behind the equality heuristic.

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$, and let $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ be a k-arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y\in\mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 4.1 (Conic-loss and uncovered-locus inequalities). *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \ge q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{10}$$

If A is $\mathcal{C}$-complete, then

$$\boxed{q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{11}$$

The first correction in (11) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Define

$$\begin{aligned} L_1(q) &= \min\left\{k \ge 3 : q^2 - k \le \binom{k}{2}(q-1)\right\}, \\ L_2(q) &= \min\left\{k \ge 3 : q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4}\right\}. \end{aligned}$$

Then

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \tag{12}$$

The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$.

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} = \begin{cases} \dfrac{k-1}{2}(k(q-1)-(k-2)(k-3)), & k \text{ even}, \\ \dfrac{k}{2}((k-1)(q-1)-(k-2)(k-3)), & k \text{ odd}. \end{cases} \tag{13}$$

For comparison, some values are

q	8	9	11	13	16	17	19
$L_1(q)$	5	5	6	6	7	7	7
$L_2(q)$	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2,16)$ has at least eight points.

Theorem 4.2 (Explicit additive lower bound)**.** *For every prime power q,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{14}$$

Consequently,

$$\liminf_{q\to\infty}(\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \tag{15}$$

Proof. Let k be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (14) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m}\binom{k}{4} \geq \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (11) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2}(k(q-1)-(k-2)(k-3)).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is exactly

$$\frac{2a-3}{4}s^3 + \frac{a^2-7a+10}{4}s^2 + \frac{-3a^2+10a-8}{2}s + \frac{-a^3+5a^2-8a+6}{2}.$$

For $0 \leq a \leq 2$, the last three coefficients are at most $5/2$, 1, and 3, respectively. Since $s \geq 2$, their total contribution is at most $4s^2$. The displayed difference is nonnegative, and hence

$$0 \leq \frac{2a-3}{4}s^3 + 4s^2.$$

Dividing by s^2 gives $a \geq 3/2 - 8/s$, which is (14). Letting q tend to infinity gives (15). $\square$

Remark 4.3 (The scale of the conic term). Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \le I_{\mathcal{C}}(A) \le 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (11) can be useful for finely balanced finite cases, but cannot improve the additive constant 3/2 in Theorem 4.2. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

5 An upper-bound transfer from complete arcs

Let $t_2(2,q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2,q)$.

Theorem 5.1 (Projective averaging). *Let $H \subseteq \mathrm{PG}(2,q)$ and let B be an ordinary complete b-arc. If*

$$b|H| < q^2 + q + 1, \tag{16}$$

then some projective image of B is disjoint from H and complete outside H.

Proof. Choose a uniformly random projectivity $g \in \mathrm{PGL}(3,q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2+q+1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g. Projectivities preserve ordinary completeness, so gB is complete outside H. □

Corollary 5.2 (Conic transfer). *If $t_2(2,q) \le q$, then*

$$\rho_{\mathcal{C}}(q) \le t_2(2,q). \tag{17}$$

More generally, every complete b-arc with $b \le q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q+1$. For integral $b \le q$ one has $b(q+1) < q^2+q+1$, and Theorem 5.1 applies. □

Kim and Vu proved that, for all sufficiently large q, every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [8]. This is less than q for large q, so Corollary 5.2 gives the following immediate consequence.

Corollary 5.3. *For some absolute constant c and all sufficiently large prime powers q,*

$$\rho_{\mathcal{C}}(q) \le \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

6 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν, and every line through ν is one of these $q+1$ tangents.

Proposition 6.1 (The nucleus belongs to the arc)**.** *Let A be a k-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ A-secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \geq k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

If A is $\mathcal{C}$-complete, then

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{18}$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k-1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν, which would place three points of A on one line. Hence these are exactly the tangent A-secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (18) follows from Corollary 4.1. ☐

Proposition 6.2 (The nucleus lies outside the arc)**.** *Let A be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

the tangent A-secants are exactly the secants through ν, and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

Consequently,

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{19}$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν. Thus the number of tangent A-secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 4.1. ☐

Corollary 6.3 (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

Consequently, inequality (11) *can always be strengthened by replacing its final term by at least $-1/m$.*

Proof. If the nucleus belongs to A, Proposition 6.1 gives $I_{\mathcal{C}}(A) \geq k-1 \geq 1$. Otherwise Proposition 6.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. ☐

These two cases are useful structural reductions, but Remark 4.3 limits what the incidence term alone can accomplish. For example, at $(q,k) = (16,8)$ inequality (11) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

7 Verified finite-field examples

The values at $q = 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 16$, the same ingredients give $8 \le \rho_{\mathcal{C}}(16) \le 9$; a checked projective classification excludes eight.

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [6] and Ball [4]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of x. The zero or nonzero status of this evaluation is independent of the chosen representative.

Lemma 7.1 (Uncovered evaluation obstruction)**.** *Let A and U be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*
2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

Proof. If f vanishes on U, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$. □

For the full system of degree-d forms, the evaluation functionals are the degree-d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

Proposition 7.2. *For the standard conic*

$$\mathcal{C} : \; XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(8) = 6, \qquad \rho_{\mathcal{C}}(9) = 6, \qquad \rho_{\mathcal{C}}(11) = 6.$$

Proof. Equation (12) gives

$$L_2(8) = L_2(9) = L_2(11) = 6.$$

Appendix A lists $\mathcal{C}$-complete arcs of size six in each plane. Their verification is described below. □

Theorem 7.3 (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let A be any eight-arc in* $\mathrm{PG}(2, 16)$*, and let $U(A)$ be the points outside A that lie on no secant of A. There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \qquad \text{and} \qquad A \cap Z(Q) = \varnothing.$$

The assertion includes singular quadratic forms.

Proof. Any four points of A form a projective frame. Normalize that frame and apply canonical augmentation. This gives respectively 4, 61, 454, and 2633 projective classes at sizes five through eight (from 182, 532, 5155, and 21495 legal extensions). Every retained transition is certified by an invertible 3×3 matrix together with pointwise projective scalar equalities, so the classification does not trust the canonical-labeling program. The final count of 2633 projective eight-arc classes independently reproduces Theorem 3.8.1 of Al-Seraji–Al-Ogali [2]; the count itself is not new.

For 2630 of the classified leaves, six points of $U(A)$ give an invertible 6×6 evaluation matrix, giving the first alternative of Lemma 7.1. In each of the remaining three leaves the evaluation rows have rank five, but an explicit linear combination places the evaluation functional of a point of A in their span, giving the second alternative.

Finally, the obstruction is projectively invariant. If gA is a normalized representative, replace Q by $Q \circ g^{-1}$. This is nonzero because g is invertible, its zero set is $gZ(Q)$, and projectivities carry secants to secants, so $U(gA) = gU(A)$. Thus a quadratic as in the statement would contradict the checked obstruction for the normalized leaf. The companion Lean development kernel-checks every local alternative; its global conic corollary additionally checks frame reduction and projective transport. ☐

Corollary 7.4 (Exact relative value over $\mathbb{F}_{16}$). *For every nonsingular conic $\mathcal{C} \subseteq \mathrm{PG}(2,16)$,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (12) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem 7.3. The nine-point witness in Appendix A gives the matching upper bound. ☐

The published ordinary classification records arc counts, stabilizers, secant-index distributions, and conic-contained classes. We did not find the uncovered-locus rank split, Theorem 7.3, or its exact relative-conic corollary tabulated there.

Remark 7.5 (Anatomy of the three exceptional leaves). Using the binary $\mathbb{F}_{16}$ encoding from Appendix A and coefficient order $(X^2, Y^2, Z^2, XY, XZ, YZ)$, the one-dimensional kernels of the three exceptional evaluation matrices are generated by

$$(1,1,1,1,1,0), \qquad (0,0,0,5,4,1), \qquad (2,1,1,5,5,1).$$

The first and third forms split over $\mathbb{F}_{16}$ as

$$(X + 6Y + 6Z)(X + 7Y + 7Z) \quad \text{and} \quad 2(X + 4Y + 15Z)(X + 15Y + 4Z),$$

respectively, and each union of lines contains two selected points. The middle form is nonsingular and contains seven of the eight selected points. Thus the exceptional leaves are not merely unexplained rank defects: two are reducible quadratic configurations, while the third is an arc differing by one point from a seven-point subset of a conic. These descriptions follow by direct field arithmetic and are not used to establish classification completeness.

For reference, the verified incidence statistics are

q	$\lvert A\rvert$	$\binom{\lvert A\rvert}{2}$	$q^2 - \lvert A\rvert$	$I_{\mathcal{C}}(A)$	$\max_{x \notin A} r(x)$
8	6	15	58	16	3
9	6	15	75	13	3
11	6	15	115	0	3
16	9	36	247	32	4

The row at $q = 11$ has a stronger geometric interpretation. Since $I_{\mathcal{C}}(A) = 0$, none of the 15 secants meets $\mathcal{C}$: every secant of the witness is exterior to the conic. If N_i denotes the number of required points of index i, relative completeness and the maximum-index bound give only $i = 1, 2, 3$, while the two moment equations give

$$N_1 + N_2 + N_3 = 115, \qquad N_1 + 2N_2 + 3N_3 = 150, \qquad N_2 + 3N_3 = 45.$$

Consequently $(N_1, N_2, N_3) = (90, 15, 10)$.

Remark 7.6 (Auxiliary residual-game structure). There is also a game-theoretic consequence of this exceptional row, not used in any bound above. After the six witness points are occupied, all twelve points of $\mathcal{C}$ remain legal. Join two conic points when their chord contains a witness point. The resulting 12-vertex, 30-edge, 5-regular conflict graph is the icosahedral graph. Its antipodal involution is fixed-point-free, preserves adjacency, and pairs only nonadjacent vertices, so the usual copycat strategy makes the corresponding independent-set building position a P-position. The companion Lean development checks every seeded continuation, proves an exact recursive localization from the projective cap game to this independent-set game, and hence proves that the displayed six-point projective position itself is P. It also checks that the displayed $q = 9$ six-arc covers every projective point, including the conic, so that position is terminal P. These observations are consequences of the displayed witnesses, not novelty claims.

The supplementary Python program `verify_relative_conic_arcs.py` performs exact finite-field arithmetic and, for each witness:

(a) enumerates and normalizes all $q^2 + q + 1$ projective points;

(b) checks that $XZ = Y^2$ contains exactly $q + 1$ points;

(c) verifies disjointness from the conic and that every pair-line contains exactly two selected points;

(d) verifies that every point outside $A \cup \mathcal{C}$ lies on a selected pair-line; and

(e) independently checks both equations in Proposition 2.2.

Its SHA-256 digest is

`e9508958d604e68c6c3d09fd3afadfaa8a3126508a51f1dfa993e7a7aed5d36a`.

The verifier is deliberately simpler than a search program: it certifies the listed upper-bound witnesses but makes no claim about search completeness.

The separate exact-classification generator `search_rhoc16.cpp` has SHA-256 digest

`589af8430e94b4c9c23ce895e6d32d2b3b5b9b387b1fb23ed6d3875cdee39031`.

Its frozen report `search_rhoc16_output.txt` has digest

`6989079b5cb64b0e57d5c42b872093fff99f861300b8fbb909daef450c15cc63`.

The generator emits local Lean certificates; its own assertions and canonical labels are not part of the trusted proof boundary.

The companion Lean library `lean/RelativeConicArcs/` formalizes the incidence, defect, conic, asymptotic, averaging, and nucleus arguments. Its generic Boolean checker inspects conic disjointness, the arc condition, and coverage of the canonical projective representatives $[1 : y : z]$, $[0 : 1 : z]$, and $[0 : 0 : 1]$; a proved normalization and soundness theorem transports acceptance to relative completeness. The concrete checks use kernel reduction rather than `native_decide`. The trust manifest `lean/RelativeConicArcs/TRUST.md` records the verifier hash, the theorem map, and axiom profiles. In particular, the finite results do not depend on the external Kim–Vu input used for the conditional asymptotic upper bound.

8 Further questions

The elementary theory leaves several concrete problems.

Problem 8.1. Determine whether the uncovered-quadratic-rank obstruction used at $q = 16$ extends to other even orders or gives useful lower bounds on the size of a $\mathcal{C}$-complete arc beyond the scalar defect inequality.

Problem 8.2. Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 5.1 currently supplies only the general complete-arc bound $O(\sqrt{q}(\log q)^c)$.

Problem 8.3. Classify equality and bounded-defect configurations in Theorem 3.1. The equality pattern requires all nontrivial secant concurrence to occur at points of maximum possible index, a severe combinatorial restriction.

Problem 8.4. Determine whether the conic stabilizer or its coherent configuration yields useful finite-order restrictions not reducible to a bound on $I_{\mathcal{C}}(A)$. Remark 4.3 shows that an estimate for $I_{\mathcal{C}}(A)$ alone cannot improve the asymptotic constant obtained here.

The central distinction is between universal overlap, already measured by the classical second index equation, and genuinely conic-specific geometry. The former gives the lower bound in this paper. Progress beyond it will probably require an explicit construction, an algebraic obstruction, or a classification of near-extremal secant arrangements rather than another application of the same incidence budget.

A Witness coordinates and field encodings

All coordinates are homogeneous and use the standard conic $XZ = Y^2$.

For $q = 8$, write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$ and encode $a_0 + a_1\alpha + a_2\alpha^2$ by the binary integer $a_0 + 2a_1 + 4a_2$. A witness is

$$(0, 1, 1), (0, 1, 2), (1, 0, 1),$$
$$(1, 0, 2), (1, 1, 2), (1, 1, 6).$$

For $q = 9$, write $\mathbb{F}_9 = \mathbb{F}_3[\alpha]/(\alpha^2 + 1)$ and encode $a_0 + a_1\alpha$ by $a_0 + 3a_1$. A witness is

$$(1, 0, 4), (1, 0, 5), (1, 1, 0),$$
$$(1, 1, 2), (1, 2, 3), (1, 2, 4).$$

For $q = 11$, coordinates lie in the prime field. A witness is

$$(1, 10, 0), (1, 9, 1), (1, 4, 7),$$
$$(1, 8, 5), (0, 1, 4), (1, 1, 7).$$

For $q = 16$, write $\mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$ and use binary coefficient encoding. A witness of size nine is

$$(1,5,14),(1,5,3),(1,9,2),(1,0,9),(1,13,13),$$
$$(1,6,10),(1,3,9),(0,1,11),(1,10,2).$$

References

[1] S. Alabdullah and J. W. P. Hirschfeld, A new lower bound for the smallest complete (k,n)-arc in $\mathrm{PG}(2,q)$, *Designs, Codes and Cryptography* **87** (2019), 679–683. doi:10.1007/s10623-018-00592-8.

[2] N. A. M. Al-Seraji and R. A. B. Al-Ogali, Classification of arcs in finite projective plane of order sixteen, *Al-Mustansiriyah Journal of Science* **29**(1) (2018), 138–149. doi:10.23851/mjs.v29i1.184.

[3] S. Ball, On small complete arcs in a finite plane, *Discrete Mathematics* **174** (1997), 29–34. doi:10.1016/S0012-365X(96)00315-9.

[4] S. Ball, On arcs and quadrics, in *Arithmetic of Finite Fields—WAIFI 2016*, Lecture Notes in Computer Science **10064**, Springer, 2017, 95–102. doi:10.1007/978-3-319-55227-9_7.

[5] D. Bartoli, A. A. Davydov, S. Marcugini, and F. Pambianco, On the smallest size of an almost complete subset of a conic in $\mathrm{PG}(2,q)$ and extendability of Reed–Solomon codes, *Problems of Information Transmission* **54** (2018), 101–115. doi:10.1134/S0032946018020011.

[6] D. G. Glynn, On the construction of arcs using quadrics, *Australasian Journal of Combinatorics* **9** (1994), 3–19.

[7] J. W. P. Hirschfeld, *Projective Geometries over Finite Fields*, second edition, Oxford University Press, 1998.

[8] J. H. Kim and V. H. Vu, Small complete arcs in projective planes, *Combinatorica* **23** (2003), 311–363. doi:10.1007/s00493-003-0024-1.

[9] Z. L. Nagy, Saturating sets in projective planes and hypergraph covers, *Discrete Mathematics* **341** (2018), 1078–1083. doi:10.1016/j.disc.2018.01.011.

[10] S.-L. Ng and P. R. Wild, On k-arcs covering a line in finite projective planes, *Ars Combinatoria* **58** (2001), 289–300.